Validity Analysis: MILU

Target context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Overall risk: **HIGH**

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM's feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content ⚠	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ⚠	1	Serious concern	high
Output Content	2	Significant gaps	medium
Output Form ⚠	1	Serious concern	high
Average	**2.2**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is the most relevant publicly available benchmark for Indic-language knowledge evaluation and provides genuine value for a portion of this deployment's needs — particularly for measuring MCQ-level accuracy on Indian Polity, History, Geography, and Economics in Hindi. Its India-first design [Q11], real-exam sourcing [Q24, Q25], and 41-subject taxonomy [Q42] are well-suited to a central-exam preparation context in principle. However, two structural issues sharply limit its validity for THIS deployment: (1) Output form/ontology mismatch (Output Ontology=1, Output Form=1) — MILU evaluates only MCQ option selection, while the deployment requires open-ended Hindi explanatory rationales; this is a fundamental scope gap acknowledged by the paper itself [Q85] and confirmed by data inspection [DATASET-D1, D9]. (2) Hindi partition content concerns (Input Content=2) — dataset analysis found 100% is_translated=True in the 26-item Hindi validation sample [DATASET-D13], heavy Engineering/Tech skew misaligned with UPSC/SSC syllabi [D5, D6, D7, D16, D36], Current Affairs staleness [D4, D25, D33], and pervasive cross-language item duplication. Input form (4) is the strongest dimension; output ontology and output form are the weakest. The benchmark can serve as a partial accuracy probe for a subset of priority subjects but cannot validate the deployment's core output.

ID	Evidence
Q11	“We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams.” (p.2)
Q24	“These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others.” (p.3)
Q25	“We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages.” (p.3)
Q42	“Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies.” (p.4)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)
D1	राष्ट्रीय आपातकाल घोषति करने के लिए 'सशस्त्र वद्रिोह' शब्द संविधान में कब जोड़ा गया? … 44वें संविधान संशोधन अधनियिम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D4	मई 2022 में इराक के सुलेमानयिा में आयोजति तीरंदाजी एशयिा कप 2022 स्टेज 2 अभयिान में भारत ने कतिने स्वर्ण पदक जीते? … आठ — Current Affairs question (2022) about archery medal count — time-bounded, may be stale
D5	जब एक डीसी सीरीज मोटर बनिा लोड के चलती है: … मोटर की गतबिहुत अधकि होती है — Engineering/Physics question in Hindi, `is_translated: True` — not in UPSC/SSC core syllabus
D6	हाफ वेव रेक्टफिायर का आउटपुट क्या होता है: … पल्सेटगि डीसी — Technical engineering question in Hindi, `is_translated: True` — domain not central to UPSC/SSC GS
D7	स्थरि धारा ट्रांसफार्मर _______ प्रकार का होता है। … शेल — Electrical engineering, translated — technical domain not prioritized for Hindi-medium UPSC prep
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest … what is the annual rate of interest? … 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition
D13	All 26 Hindi examples: is_translated = True — Every Hindi validation example in the sample is machine-translated from English
D16	फॉरट्रान 77 के फक्सि्ड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? … एक जंप लेबल या फॉर्मेट लेबल — Computer Science / Fortran programming question in Hindi — very niche for UPSC/SSC aspirants
D25	ਫਰਵਰੀ - ਮਾਰਚ 2022 ਵਿੱਚ ਹੋਈ ਰਾਜ ਵਧਿਾਨਸਭਾ ਚੋਣਾਂ ਵਿੱਚ, ਆਮ ਆਦਮੀ ਪਾਰਟੀ (AAP) ਨੇ … ਪੰਜਾਬ — Current Affairs (2022 state elections) in Punjabi — politically specific, time-bounded
D33	ফব্রেুয়ারি- মার্চ ২০২২ এ অনুষ্ঠতি রাজ্য বধিানসভা নর্বিাচনে, আম আদমি পার্টি(AAP) … পাঞ্জাব — 2022 state elections Current Affairs — AAP, UP, Goa all named; time-bounded content
D36	एएम की बैंडवड्थि _________ है। … 1110 kHz — AM bandwidth engineering question — highly technical, `is_translated: True`; unlikely in UPSC GS

Practical Guidance

What This Benchmark Measures

MILU measures MCQ-label accuracy on Hindi-language Indian-domain knowledge across 8 macro-domains and 41 subjects. For this deployment, it provides moderate evidence for a model's ability to produce correct verdicts on UPSC/SSC-style Polity, History, Geography, and Economics MCQs in Hindi (Input Ontology partial coverage, Input Form strong, Input Content partial). It offers no evidence on the deployment's core output: the Hindi explanatory rationale (Output Ontology and Output Form both score 1).

Construct Depth

Construct depth is shallow for this deployment. MILU probes whether a model can rank the correct option among up to four — a necessary but far-from-sufficient signal for exam-feedback quality. It does not probe pedagogical clarity, factual rationale accuracy, fluency, register appropriateness, or robustness to authentic Hindi-medium exam phrasing. Combined with cross-language duplication, possible high translation rate in Hindi, and Current Affairs staleness, even the verdict-accuracy signal is partially construct-irrelevant for 2025–26 central-exam readiness.

What Else You Need

Three supplementations are essential: (1) A custom Hindi-language rationale-quality evaluation framework with rubric covering factual accuracy, fluency, pedagogical clarity, and register — no public resource exists [WEB-13, WEB-14]. (2) A central-exam-specific subset of MILU filtered to UPSC/SSC/banking items with explicit Mathematics/Reasoning and Current Affairs categories, ideally drawing on natively authored Hindi items rather than translated content. (3) A current-affairs evaluation set refreshed annually with 2024–2026 events to address staleness.

ID	Evidence
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)